Title of Subject	:	**Hydro-informatics: Data Management and Analysis (2+1)**					
Course Code	:	**HID-661**					
Semester	:	Second (2nd)		Year : First (1st)			
Discipline	:	Hydraulics, Irrigation and Drainage (HID)					
Effective	:	2026 Batch and onwards					
Assessment	:	**Theory (50%)** 20% Test/Assignment/Quiz 20% Mid-Semester Examination 60% Final Written Examination **Lab (50%)** 20% Lab Quiz 20% Lab Evaluation 60% Lab Exam					
Credit Hours	:	**Theory**	02	**Practical**	01	**Total**	03
Contact Hours	:	**Theory**	28	**Practical**	42	**Total**	-
Marks	:	**Theory**	50	**Practical**	50	**Total**	100

1. Course Description

Hydroinformatics integrates hydrological science, geospatial technologies, computational methods, and data analysis for water resources management. The course introduces hydrological data sources and analysis, GIS, remote sensing, hydrological and hydraulic modeling, machine learning, and applications in flood, drought, and water quality management.

2. Course Objectives

- Understand the principles and scope of Hydroinformatics.
- Acquire, manage, and analyze hydro-meteorological and geospatial data.
- Apply statistical, GIS, remote sensing, modeling, and introductory machine learning tools to water resources problems.
- Explore Hydroinformatics applications in flood, drought, and water quality management.
- Develop basic Hydroinformatics workflows for decision support and communicate technical findings.

3. Learning Outcomes

By the end of the course, students will be able to:

- Identify and use relevant hydro-meteorological and geospatial datasets.
- Perform basic statistical and time-series analysis of hydrological data.
- Apply GIS and remote sensing tools to water resources problems.
- Develop basic rainfall-runoff and hydraulic modeling workflows.
- Apply introductory machine learning for streamflow prediction.
- Use Hydroinformatics tools for flood, drought, and water quality assessment.
- Present and document their technical work effectively.

4. Weekly Teaching and Laboratory Schedule

Week	Theory	Laboratory
1	Foundations of Hydroinformatics: A Refresher	Software setup & datasets
2	Introduction to Hydroinformatics	Cloud Computing using Google Earth Engine
3	Statistics Refresher	Python statistics & visualization
4	Hydrological Data Sources	WaPOR, Pakistan Drought Management System (PakDMS), GRACE, Earth Console, etc.
5	GIS for Hydroinformatics	Watershed delineation
6	Hydrological Time Series	Hydrographs & trends
7	Hydrological Modeling	Rainfall-runoff modeling
8	Midterm	-
9	Field Visit	Reflection and Field Report
10	Machine Learning	Streamflow prediction
11	Hydraulic Modeling	Introduction to RAS Mapper
12	Flood Hydroinformatics	Flood mapping using SAR data and GEE
13	Drought Hydroinformatics	SPI & drought mapping
14	Water Quality Informatics	Water quality analysis
15	Decision Support Systems	Project workshop
16	Project Presentations	Presentation

5. Suggested Semester Project Areas

- Rainfall-runoff modeling
- Flood susceptibility mapping
- Drought monitoring
- Streamflow forecasting
- Water quality analysis
- Hydroinformatics dashboard

- WaPOR-based evapotranspiration and water productivity
- Satellite-based water level monitoring

6. Software

Python, Jupyter Notebook, QGIS, ArcGIS Pro (if available), Google Earth Engine, HEC-HMS, RAS Mapper (demonstration), Excel.

7. Open Data Sources

FAO WaPOR, CHIRPS, ERA5, NASA POWER, GPM IMERG, Sentinel Missions, Landsat, MODIS, SMAP, HydroSHEDS, JRC Global Surface Water, ESA WorldCover, PMD (where available), SIDA, etc.

8. Recommended Books and Reading Materials:

1. Kumar et al., "Hydro-informatics: Data Integrative Approaches in Computation, Analysis, and Modeling", CRC Press, (latest edition)
2. Tomer, S.K., Python in Hydrology, Green Tea Press, Indian Institute of Science, (latest edition), http://www.greenteapress.com/pythonhydro/pythonhydro.html
3. Desai, P. Python Programming for Arduino. Packt Publishing Ltd. (latest edition).
4. Géron, A. "Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems", O'Reilly Media, Inc., (latest edition).
5. Remote Sensing and Image Interpretation, Lillesand et al.

9. Suggested Journals

1. Journal of Hydroinformatics
2. Journal of Hydrology

Approval		
Board of Studies, USPCASW	Res. No.	Dated:
Board of Faculty of A&CE	Res. No.	Dated:
Advanced Studies and Research Board	Res. No.	Dated:
Academic Council	Res. No.	Dated: